

Project Approach (Roadmap) – Batch 7

Machine Learning-Based Healthcare Data Analytics for Early Disease Risk Prediction

Domain

Healthcare Analytics

Healthcare Analytics is the process of collecting, analyzing, and interpreting patient healthcare data to improve medical decision-making. In this project, we use Machine Learning to analyze patient health records and predict the risk of diseases such as diabetes or heart disease at an early stage.

Dataset

1. UCI Heart Disease Dataset –

Contains:

- Age
- Gender
- Chest Pain Type
- Cholesterol
- Blood Pressure
- ECG Results
- Maximum Heart Rate
- Exercise-induced Angina
- Target (Disease / No Disease)

<https://archive.ics.uci.edu/dataset/45/heart+disease>

2. Diabetes Prediction Dataset (Kaggle) –

Contains patient information such as:

- Pregnancies
- Glucose Level
- Blood Pressure
- Skin Thickness
- Insulin
- BMI
- Diabetes Pedigree Function
- Age
- Outcome (0 = No Diabetes, 1 = Diabetes)

<https://www.kaggle.com/datasets/iammustafatz/diabetes-prediction-dataset>

3. UCI Chronic Kidney Disease Dataset -

Contains:

- Blood Pressure
- Specific Gravity
- Albumin
- Sugar

- Blood Glucose
- Blood Urea
- Serum Creatinine
- Hemoglobin
- Target

<https://archive.ics.uci.edu/dataset/336/chronic+kidney+disease>

Abstract

Healthcare organizations generate a vast amount of patient data every day, including medical history, laboratory test results, demographic information, and clinical measurements. Analyzing this data manually is time-consuming and may delay the early detection of diseases. Early diagnosis is essential for improving treatment outcomes, reducing healthcare costs, and enabling timely medical intervention. Therefore, there is a need for an intelligent system that can analyze healthcare data efficiently and accurately predict disease risk.

This project proposes a Machine Learning-Based Healthcare Data Analytics System for Early Disease Risk Prediction. The system utilizes benchmark healthcare datasets containing patient attributes such as age, gender, glucose level, blood pressure, BMI, insulin level, cholesterol, and other clinical parameters. The collected data undergoes several preprocessing steps, including missing value handling, duplicate removal, encoding of categorical features, normalization/standardization, outlier treatment, and class balancing using SMOTE, to improve data quality and ensure reliable model performance.

After preprocessing, Exploratory Data Analysis (EDA) is performed to understand the dataset through descriptive statistics and visualizations such as histograms, boxplots, heatmaps, disease distribution charts, and correlation analysis. Feature engineering and feature selection techniques are then applied to identify the most influential healthcare parameters that contribute to disease prediction while reducing irrelevant or redundant features.

The processed dataset is used to train and compare multiple machine learning algorithms, including Logistic Regression, Decision Tree, Random Forest, XGBoost, and CatBoost. Each model is evaluated using standard performance metrics such as Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix. The best-performing model is selected based on its predictive performance and generalization capability.

To enhance transparency and trust in the prediction process, the project incorporates Explainable Artificial Intelligence (XAI) techniques, specifically SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations).

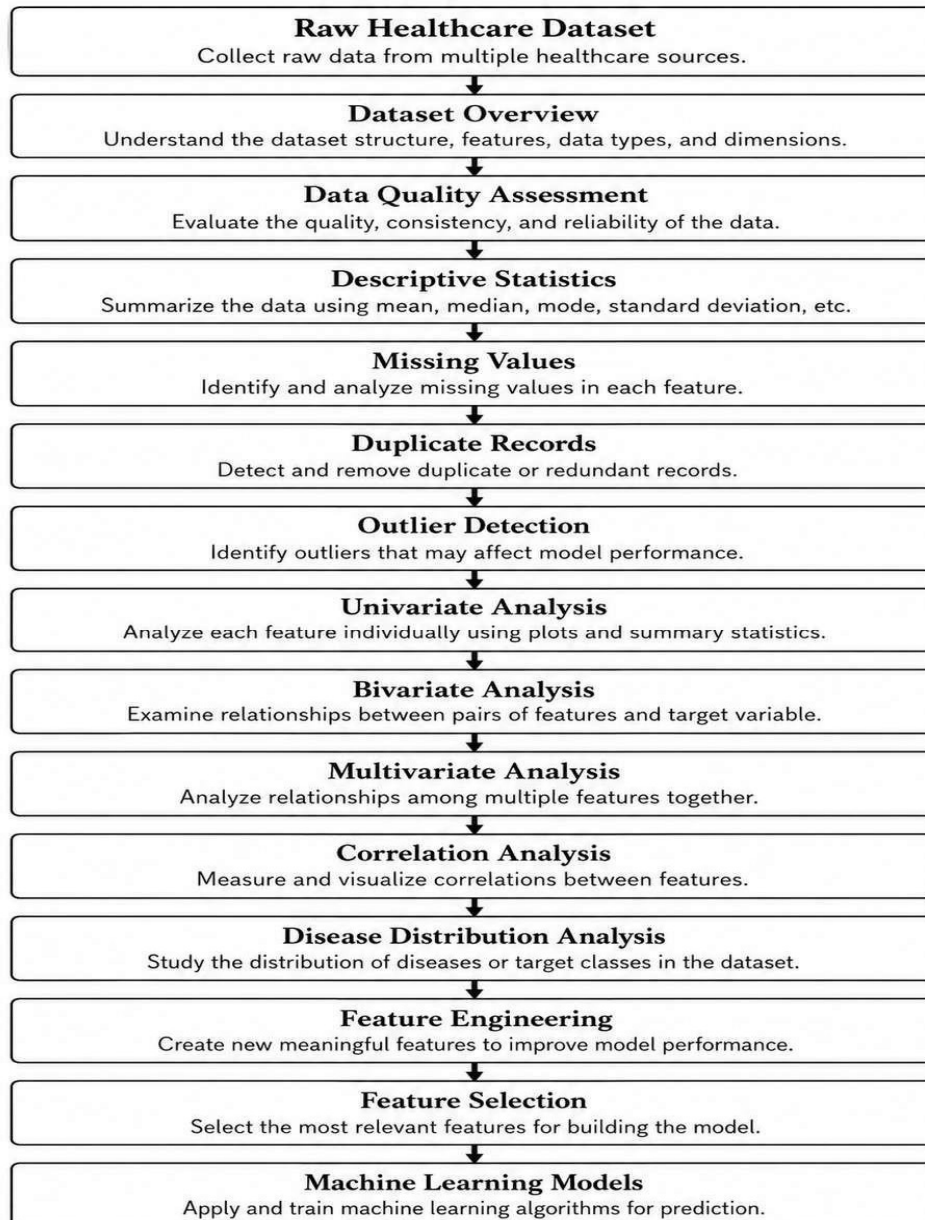
These methods explain how individual patient features influence disease predictions, enabling healthcare professionals to understand the reasoning behind the model's decisions.

Finally, the trained model is integrated into a user-friendly web application developed using Flask, where users can enter patient health information and receive real-time disease risk predictions. The prediction history is stored in a MySQL database, and an interactive dashboard presents healthcare analytics, feature importance, disease trends, and model performance through intuitive visualizations.

The proposed system aims to assist healthcare professionals by providing an accurate, interpretable, and efficient decision-support tool for early disease risk prediction, enabling timely diagnosis, personalized treatment planning, and improved patient care.

Data Analytics

EDA Flow Diagram



Preprocessing

- **Missing Value Handling:** Fill or remove missing patient health values (such as BMI, glucose, or blood pressure) to ensure accurate disease risk prediction.
- **Duplicate Removal:** Remove duplicate patient records to prevent repeated data from affecting the model's learning and prediction results.
- **Encoding:** Convert categorical patient information (such as gender or smoking status) into numerical values so the machine learning model can process them.
- **Normalization / Standardization:** Scale patient health parameters (such as glucose, BMI, and blood pressure) to a common range so that all features contribute equally during model training.
- **Outlier Treatment:** Detect and handle abnormal patient values that could negatively affect the accuracy of disease risk prediction.
- **SMOTE (Class Balancing):** Balance the dataset by generating additional samples for underrepresented disease cases, enabling the model to predict both healthy and diseased patients more accurately.
- **Train-Test Split:** Split the healthcare dataset into training and testing sets to train the model and evaluate its performance on unseen patient data.

ML Models

- **Logistic Regression:** Predicts whether a patient is at risk (Disease/No Disease) based on health parameters and serves as a simple baseline classification model.
- **Decision Tree:** Classifies disease risk by learning decision rules from patient features such as glucose level, BMI, age, and blood pressure, making predictions easy to interpret.
- **Random Forest:** Combines multiple decision trees to improve the accuracy and reliability of disease risk prediction while reducing overfitting.
- **XGBoost:** Uses gradient boosting to accurately identify complex patterns in healthcare data, resulting in improved disease prediction performance.
- **CatBoost:** Efficiently handles categorical healthcare features with minimal preprocessing and provides robust and accurate disease risk predictions.

Explainability

- **SHAP (SHapley Additive exPlanations):** Explains how each patient health feature (such as glucose, BMI, and age) contributes to the predicted disease risk, making the model transparent.
- **LIME (Local Interpretable Model-Agnostic Explanations):** Explains the prediction for an individual patient by identifying the specific health factors that influenced the disease risk.

Visualization

- **Heatmap:** Displays the correlation between healthcare features to identify the factors most strongly associated with disease risk.

- **Disease Distribution Charts:** Shows the distribution of healthy and diseased patients to understand the dataset balance.
- **Boxplot:** Detects outliers in healthcare features such as glucose level, BMI, and blood pressure.
- **Histogram:** Shows the distribution of individual patient features, such as age, BMI, or glucose levels.
- **ROC Curve:** Evaluates the model's ability to distinguish between diseased and healthy patients by measuring classification performance.
- **Confusion Matrix:** Displays correct and incorrect disease predictions, helping evaluate the model's classification accuracy.
- **Feature Importance:** Identifies the healthcare features that have the greatest impact on disease risk prediction.
- **SHAP Summary Plot:** Visualizes the overall contribution of all healthcare features to the model's predictions across the entire dataset.
- **Dashboard:** Presents patient statistics, prediction results, disease trends, and model performance in an interactive interface for easy analysis.

Comparative Analysis

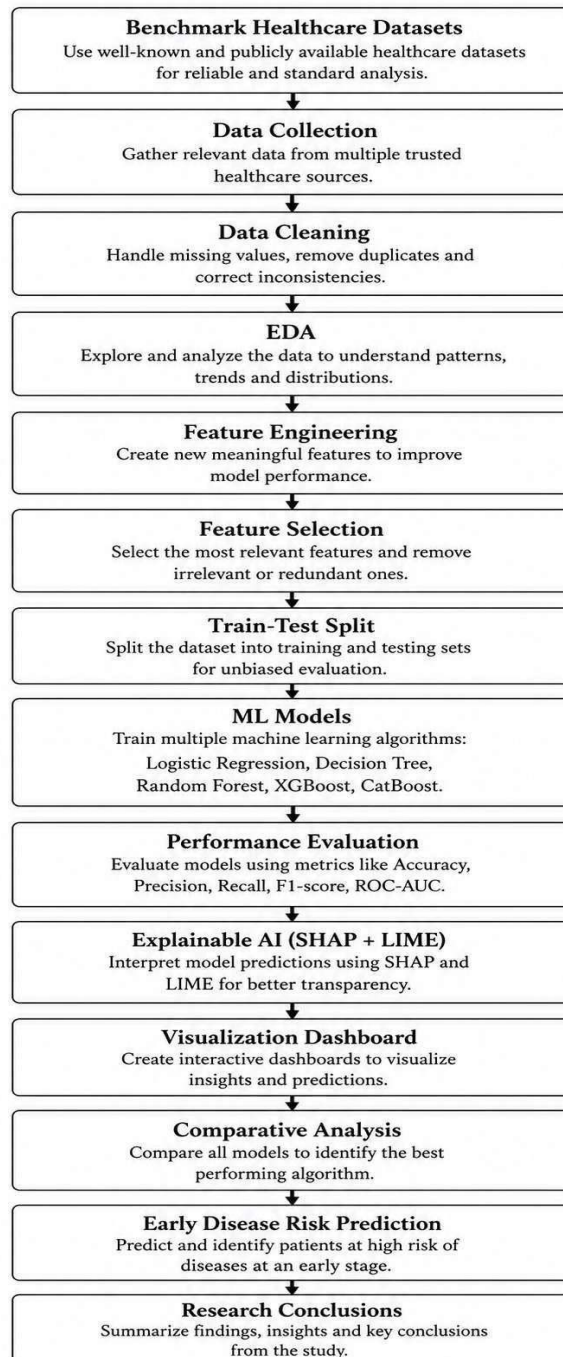
Compare:

- **Logistic Regression:** Serves as the baseline model to compare disease prediction performance with other machine learning algorithms.
- **Decision Tree:** Evaluates how well a rule-based model predicts disease risk using patient health parameters.
- **Random Forest:** Compares the performance of an ensemble model to determine if it improves prediction accuracy and reduces overfitting.
- **XGBoost:** Measures the effectiveness of gradient boosting in identifying complex disease patterns from healthcare data.
- **CatBoost:** Evaluates the model's ability to accurately predict disease risk, especially when handling categorical healthcare features.

Metrics:

- **Accuracy:** Measures the overall percentage of correctly predicted disease and non-disease cases.
- **Precision:** Measures how many patients predicted as diseased are actually diseased.
- **Recall (Sensitivity):** Measures how effectively the model identifies patients who actually have the disease.
- **F1-Score:** Provides a balanced evaluation of Precision and Recall for disease prediction.
- **ROC-AUC:** Evaluates the model's ability to distinguish between diseased and healthy patients across different thresholds.
- **Confusion Matrix:** Displays the number of correct and incorrect disease predictions made by each machine learning model.

Recommended Research Pipeline



BATCH:7

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